B.Tech Computer Science and Engineering Syllabus 2026 (Sample University)

Second Year Core Courses

SAMP-MA201 Probability and Statistics. 4 credits. Semester 3. Prerequisite: SAMP-MA101 Calculus. Topics: random variables, distributions, estimation, hypothesis testing, regression.

SAMP-CS201 Data Structures. 4 credits. Semester 3. Prerequisite: SAMP-CS101 Programming in Python. Topics: arrays, linked lists, trees, graphs, hashing, complexity analysis.

SAMP-CS202 Operating Systems. 3 credits. Semester 4. Prerequisite: SAMP-CS201 Data Structures.

Third Year Core Courses

SAMP-CS301 Machine Learning. 4 credits. Semester 5. Prerequisite: SAMP-MA201 Probability and Statistics and SAMP-CS201 Data Structures. Topics: supervised learning, model evaluation, decision trees, support vector machines, clustering, neural network basics.

SAMP-CS305 Database Systems. 3 credits. Semester 5. Prerequisite: SAMP-CS201 Data Structures. Topics: relational model, SQL, normalization, transactions, indexing.

SAMP-CS310 Computer Networks. 3 credits. Semester 6. Prerequisite: SAMP-CS202 Operating Systems.

Fourth Year Courses

SAMP-CS401 Deep Learning. 4 credits. Semester 7. Prerequisite: SAMP-CS301 Machine Learning. Topics: convolutional networks, sequence models, transformers, training at scale.

SAMP-CS499 Capstone Project. 8 credits. Semester 8. Prerequisite: completion of at least 120 credits.

Total credits required for the B.Tech CSE degree: 160.